CMR UNIVERSITY
Private University Estd in Karnataka State by Act No. 45 of 2013

SCHOOL OF ENGINEERING AND TECHNOLOGY

A Project Report
On

"Wild Mushroom Toxicity Identification & Poison Prevention"

Submitted in partial fulfillment of the requirements for the award of degree in

BACHELOR OF TECHNOLOGY
IN
Computer Engineering

SUBMITTED BY-
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SUBJECT-
Machine Learning and Applications

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2026-2027

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Chagalahatti, Bengaluru, Karnataka-562149

Department of Computer Engineering

CERTIFICATE

This is to certify that the Project entitled "**Wild Mushroom Toxicity Identification & Poison Prevention** " has been successfully carried out by Suhas V (24BBTCO066) in partial fulfillment of the requirement for the award of the degree **Bachelor of Technology in Computer Engineering of CMR University,** Bengaluru during the academic year **2026-2027.**The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide

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Dr. Sudarshan Babu Pandava
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Signature of HOD

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Finally, I would like to express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

Suhas V-(24BBTCO066)

DECLARATION

I, Suhas V (24BBTCO066), a student of 5th Semester B.Tech, Computer Engineering, School of Engineering and Technology, CMR University, Bengaluru, hereby declare that the project work entitled **"Wild Mushroom Toxicity Identification & Poison Prevention"** has been carried out by me under the guidance of **Dr. Sudarshan Babu Pandava**, Department of Computer Engineering, School of Engineering and Technology.

I further declare that this report is submitted in partial fulfillment of the requirements for the award of the Bachelor of Technology (B.Tech.) in Computer Engineering by CMR University, Bengaluru, during the academic year 2026–2027.

I certify that this work is original and has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or other academic qualification. All sources of information used in this report have been duly acknowledged and cited.

I understand that any violation of academic integrity or plagiarism may result in disciplinary action as per the regulations of CMR University.

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Signature of the Student
Suhas V
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ABSTRACT

Wild mushroom foraging is hazardous because poisonous species often closely resemble edible ones, and a single misidentification can be fatal. This project builds an automated classifier that labels a specimen as edible or poisonous from 22 categorical macroscopic traits of the UCI Mushroom dataset, which contains 8,124 specimens. The study covers CART decision trees with Gini impurity, minimal cost-complexity pruning, a cost-sensitive tree variant, and Bernoulli Naive Bayes as a probabilistic baseline.

The data are audited and one-hot encoded into 117 binary features, and the missing stalk-root values are kept as their own category rather than imputed. An unconstrained tree reaches 100% training and test accuracy with 14 leaves and a depth of 7, and pruning keeps accuracy above 99% down to about 8 leaves before it collapses. Cross-validation, however, shows that standard unweighted trees still make about 2 poisonous-called-edible errors across the training folds.

Because calling a poisonous mushroom edible is far costlier than discarding an edible one, a cost-sensitive tree that weights the poisonous class 50 times is also evaluated. It removes these cross-validation false negatives and gives a 12-leaf tree with 0 false negatives and 0 false positives on the held-out test set. Bernoulli Naive Bayes reaches only 93.3% accuracy with 96 false negatives, and lowering its decision threshold does not close the gap. Odor is the dominant predictor in every tree fitted.

The report presents the problem formulation, dataset preprocessing, mathematical background, experimental results, a visualization gallery, and an error analysis with practical recommendations, showing how interpretable, cost-aware decision trees can support safer mushroom identification.

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Google Colab Link: https://colab.research.google.com/drive/1gddG0rBzkUWIu-TN8mUV7290j8dZpaqB?usp=sharing

Section A: Abstract & Problem Formulation

Wild mushroom foraging is hazardous because poisonous species imitate edible ones. This project builds an automated classifier that labels a specimen edible or poisonous from 22 categorical macroscopic traits of the UCI Mushroom dataset (8,124 specimens). The analysis covers CART with Gini impurity, minimal cost-complexity pruning, a cost-sensitive variant, and Bernoulli Naive Bayes as a probabilistic baseline.

An unconstrained tree reaches 100% training and test accuracy with 14 leaves and depth 7. Pruning removes leaves cheaply down to about 8 leaves (accuracy stays above 99%) and collapses below 5 leaves. Cross-validation, however, shows that standard (unweighted) trees make about 2 poisonous-called-edible errors across the training folds. Weighting the poisonous class 50 times removes these (CV false negatives 0) and gives a 12-leaf tree with 0 false negatives and 0 false positives on the held-out test set. Bernoulli Naive Bayes reaches only 93.3% accuracy with 96 false negatives, and lowering its decision threshold does not close the gap.

Problem formulation. Let $x \in \{0,1\}^{117}$ be the one-hot encoding of a specimen and $y \in \{0,1\}$ with $y = 1$ for poisonous. The classifier $f(x)$ must minimise expected cost with asymmetric losses: a false negative (poisonous called edible, lethal) is far more costly than a false positive (edible discarded). We take $c_{FN} : c_{FP} = 100 : 1$ for cost reporting. Positive class = poisonous throughout.

Section B: Dataset Exploration & Preprocessing

B.1 Data audit

The file contains 8,124 rows and 23 columns (22 attributes plus the class). The class is nearly balanced: 4,208 edible (51.8%) and 3,916 poisonous (48.2%), so accuracy is not distorted by imbalance, but recall on the poisonous class is still the key safety metric.

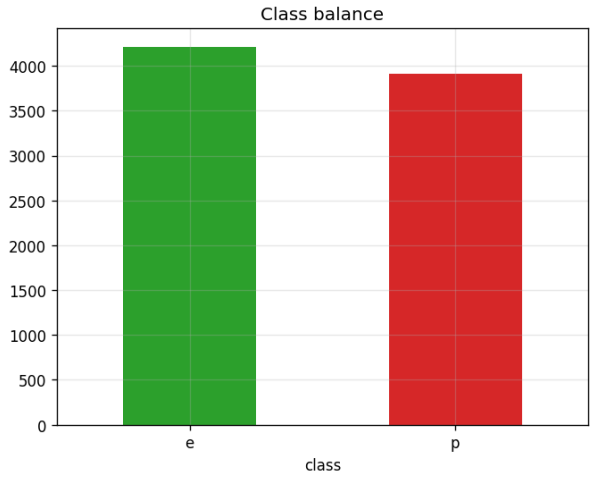


Figure 1. Class balance: 4,208 edible (e) and 3,916 poisonous (p) specimens.

Table 1. Number of categorical levels per attribute (before encoding).

Attribute	Levels	Attribute	Levels
cap-shape	6	stalk-surface-above-ring	4
cap-surface	4	stalk-surface-below-ring	4
cap-color	10	stalk-color-above-ring	9
bruises	2	stalk-color-below-ring	9
odor	9	veil-type	1
gill-attachment	2	veil-color	4
gill-spacing	2	ring-number	3
gill-size	2	ring-type	5
gill-color	12	spore-print-color	9
stalk-shape	2	population	6
stalk-root	5	habitat	7

Missing values. Only stalk-root contains missing entries, coded "?": 2,480 of 8,124 rows (30.5%). Rather than impute, we keep "?" as its own level (stalk-root_?). In this dataset the missingness pattern is itself informative about species, and imputation would invent values for nearly a third of the rows. **Constant attribute.** veil-type has a single level (p) in every row and carries no information; it encodes to one constant column that no split can use.

B.2 Encoding and split

All 22 attributes are one-hot encoded with pandas get_dummies, producing a 8,124 × 117 binary matrix with 81.2% zeros (sparse). One-hot encoding is appropriate because the levels are nominal with no order, and each binary column gives a CART split of the form "attribute = level". Data are split 80/20 with stratification on the class and random_state = 42: 6,499 training rows and 1,625

test rows (842 edible, 783 poisonous). The test set is untouched during model and hyperparameter selection, which uses stratified 5-fold cross-validation (shuffle, seed 42) on the training rows.

Section C: Mathematical Formulation

C.1 Impurity measures and split criterion

For a node t with class proportions p_k, CART uses the Gini impurity; entropy is the alternative studied here:

$$G(t) = 1 - \Sigma_k\, p_k^2 \qquad H(t) = -\,\Sigma_k\, p_k\, \log_2 p_k$$

A binary split s sends n_L samples left and n_R right (n = n_L + n_R). CART chooses the split with the largest impurity decrease:

$$\Delta G(s,t) = G(t) - (n_L/n)\, G(t_L) - (n_R/n)\, G(t_R)$$

A pure node has G = 0. Because the classes are separable on this dataset, the unconstrained tree keeps splitting until every leaf is pure, which gives zero training error.

C.2 Minimal cost-complexity pruning

For a tree T with leaf set $\tilde{T}$, the penalised risk is

$$R_\alpha(T) = R(T) + \alpha\, |\tilde{T}|, \qquad R(T) = \Sigma_\{t \in \tilde{T}\}\, r(t)\, p(t)$$

For an internal node t with subtree T_t, the effective alpha at which collapsing T_t into a leaf becomes optimal (weakest link) is

$$\alpha_eff(t) = (R(t) - R(T_t)) / (|\tilde{T}_t| - 1)$$

Repeatedly pruning the node with the smallest α_eff yields a nested sequence of subtrees $T_0 \supset T_1 \supset \ldots \supset \{root\}$, with its alpha values (sklearn: cost_complexity_pruning_path). Larger α means a smaller tree, higher bias and lower variance. Pre-pruning (max_depth, min_samples_split, min_samples_leaf) instead stops growth early.

C.3 Cost-sensitive trees

With class weights w_k, node proportions become p_k = w_k n_k / Σ_j w_j n_j, so impurity and leaf labels favour the costly class. We use w = (1, w_pois). A leaf is labelled poisonous whenever w_pois n_pois ≥ n_edible. At w_pois = 50, a leaf with only 2% poisonous training samples is already labelled poisonous.

C.4 Bernoulli Naive Bayes

With binary features x_j and class y, Bernoulli NB assumes conditional independence:

$$P(y \mid x) \propto P(y)\, \Pi_j\, \theta_{jy}\text{^}\{x_j\}\, (1 - \theta_{jy})\text{^}\{1-x_j\}, \qquad \theta_{jy} = (N_\{jy\} + \alpha) / (N_y + 2\alpha)$$

with Laplace smoothing α = 1. The independence assumption is the main weakness here: attributes such as stalk-color-above-ring and stalk-color-below-ring are strongly correlated. Under asymmetric costs the Bayes-optimal rule predicts poisonous when

$$P(\text{poisonous} \mid x) \geq \tau = c_{FP} / (c_{FP} + c_{FN}) = 1/101 \approx 0.0099 \quad (\text{for } c_{FN} : c_{FP} = 100 : 1)$$

C.5 Algorithmic pseudocode

```
CART(D, class weights w):
  if node is pure or no split reduces impurity:
      return leaf(weighted majority class)
  for each binary feature j:  compute weighted ΔGini of split (x_j = 1)
  j* = argmax ΔGini
  return node(j*, CART(D[x_j* = 0]), CART(D[x_j* = 1]))

PRUNE(T):                          # minimal cost-complexity pruning
  while T has internal nodes:
      compute alpha_eff(t) for every internal node t
      t* = argmin alpha_eff(t);  collapse subtree T_t* to a leaf
      record (alpha_eff(t*), T)
  choose alpha by 5-fold CV: fewest false negatives, then fewest leaves
```

Section D: Experimental Results & Metrics

D.1 Unconstrained tree and splitting criterion

The unpruned Gini tree fits the training data perfectly (accuracy 1.0) and also scores 1.0 on the test set, with 14 leaves and depth 7. Zero training error is expected here: the data are separable by categorical attributes, so growth stops when all leaves are pure. It does not by itself indicate overfitting; the CV and test curves must confirm it. Entropy gives an almost identical tree.

Table 2. Gini vs entropy (unpruned, test set).

Criterion	Leaves	Depth	Train acc	Test acc
Gini	14	7	1.0000	1.0000
Entropy	13	7	1.0000	1.0000

D.2 Post-pruning (cost-complexity path)

Table 3. Pruning path of the unweighted Gini tree (accuracy per effective alpha).

ccp_alpha	Leaves	Depth	Train acc	5-fold CV acc	Test acc
0.000000	14	7	1.0000	0.9997	1.0000
0.000154	12	5	0.9998	0.9997	0.9988
0.000307	11	5	0.9997	0.9997	0.9988
0.000921	10	5	0.9992	0.9992	0.9982
0.002662	9	5	0.9974	0.9977	0.9957
0.006736	8	5	0.9945	0.9951	0.9926

ccp_alpha	Leaves	Depth	Train acc	5-fold CV acc	Test acc
0.007797	7	5	0.9903	0.9917	0.9871
0.010948	6	4	0.9848	0.9829	0.9797
0.011278	5	3	0.9789	0.9829	0.9735
0.017892	4	3	0.9695	0.9734	0.9668
0.042464	3	2	0.9461	0.9588	0.9422
0.089546	2	1	0.8871	0.9200	0.8855

Reading the table: pruning from 14 to 8 leaves costs under 0.6 percentage points of test accuracy; below 5 leaves accuracy drops sharply (97.9% at 5 leaves, 88.6% at 2 leaves). Train, CV and test accuracy stay close together all along the path, so there is no sign of large variance. Mild pruning gives a smaller tree at a very small accuracy cost, but no pruned tree matches the full tree's 100% test accuracy (the simplest tree with 100% test accuracy is the unpruned 14-leaf tree).

D.3 Cost-sensitive selection

The unweighted trees show 2 false negatives under 5-fold cross-validation at every alpha tested. We therefore swept the poisonous class weight $w \in \{1, 5, 10, 50\}$ and, for each weight, every ccp alpha, choosing the model by cross-validated false negatives, then false positives, then leaf count. The test set was not used for this choice.

Table 4. Minimum 5-fold CV false negatives (training folds, n = 3,133 poisonous) by class weight.

Poisonous weight w	1	5	10	50
Min CV false negatives	2	2	2	0

Only w = 50 removes the cross-validated false negatives. The selected model has w = 50, alpha = 0 (no post-pruning), 12 leaves, depth 7, with CV false negatives 0 and CV false positives 0. At this weight pruning no longer helps: it simply trades away poisonous recall.

Table 5. Pruning strategies compared (unweighted trees unless stated).

Strategy	Leaves	Depth	CV acc	Test acc
No pruning (CART)	14	7	0.9997	1.0000
Post-pruning, alpha = 0.000307	11	5	0.9997	0.9988 (2 FN)
Pre-pruning (best grid: max_depth 5)	–	5	0.9997	0.9988
Cost-sensitive, w = 50 (selected)	12	7	1.0000	1.0000 (0 FN)

D.4 Model comparison on the test set

Table 6. Final comparison (poisonous = positive class, 1,625 test specimens).

Model	Test acc	Recall (poisonous)	FN	FP	5-fold CV acc
Unpruned tree (Gini)	1.0000	1.0000	0	0	0.9997

Model	Test acc	Recall (poisonous)	FN	FP	5-fold CV acc
Cost-sensitive tree (w = 50)	1.0000	1.0000	0	0	1.0000
Bernoulli Naive Bayes	0.9329	0.8774	96	13	0.9408

D.5 Asymmetric risk and Naive Bayes thresholding

Lowering the Naive Bayes decision threshold reduces false negatives at the price of more false positives, and it never reaches zero. Weighted cost uses c_FN : c_FP = 100 : 1 (cost = 100·FN + FP).

Table 7. Bernoulli NB: effect of lowering the poisonous probability threshold.

Threshold τ	FN	FP	Weighted cost (100·FN + FP)
0.50 (default)	96	13	9,613
0.20	75	18	7,518
0.05	46	20	4,620
0.01 (≈ Bayes-optimal)	31	24	3,124
Cost-sensitive tree	0	0	0

Even at the Bayes-optimal threshold, NB leaves 31 poisonous specimens classed as edible. Calibrated probabilities from Bernoulli NB are distorted by the correlated attributes, so the threshold shifts only part of the error.

Section E: Visualization Gallery

Figures 2 to 6 are the notebook outputs (300 dpi). Interpretations are given in the captions.

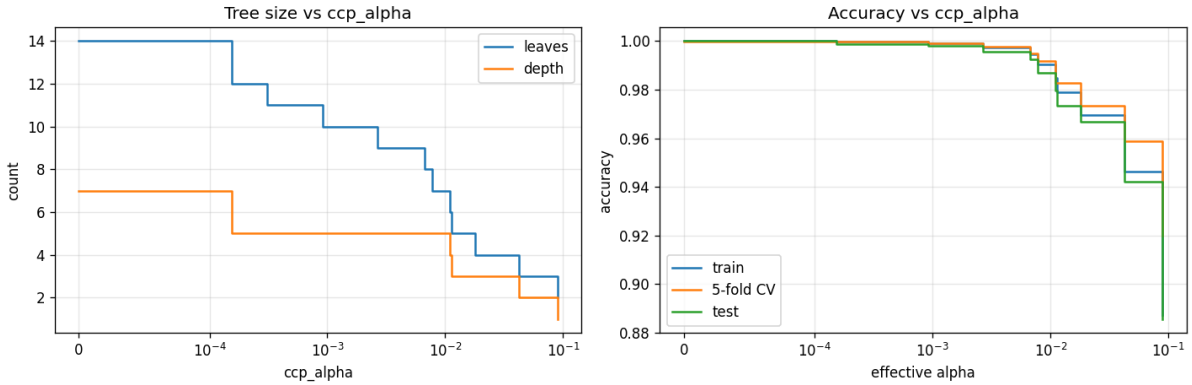


Figure 2. Left: number of leaves and tree depth versus ccp_alpha (symlog axis). Leaves fall from 14 to 2 as alpha rises, and depth drops from 7 to 5 at the first pruning step. Right: train, 5-fold CV and test accuracy versus effective alpha. Accuracy stays above 99% until about 8 leaves (alpha ≈ 0.007) and degrades rapidly beyond alpha ≈ 0.01, to 88.6% test accuracy at 2 leaves.

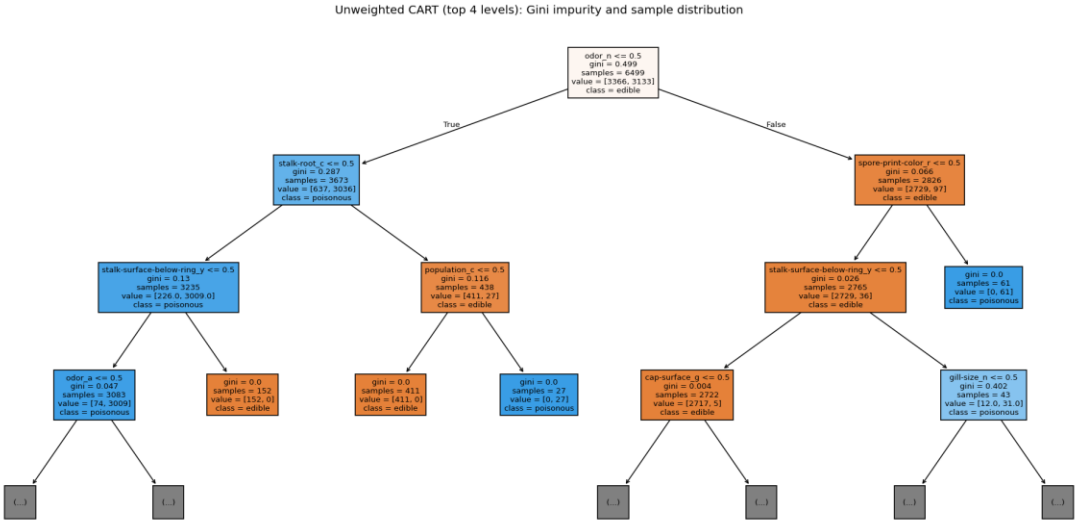


Figure 3. Unweighted CART, top 4 levels, with Gini impurity and class counts [edible, poisonous]. The root (Gini 0.499, [3366, 3133]) splits on odor = none. The odor ≠ none branch (3,673 samples, [637, 3036]) is mostly poisonous; stalk-root, stalk-surface-below-ring, population and odor = almond separate its 637 edible specimens. The odor = none branch (2,826 samples, [2729, 97]) is mostly edible; spore-print-color = r isolates a pure poisonous leaf of 61 samples, and stalk-surface-below-ring, cap-surface and gill-size separate the remaining poisonous cases.

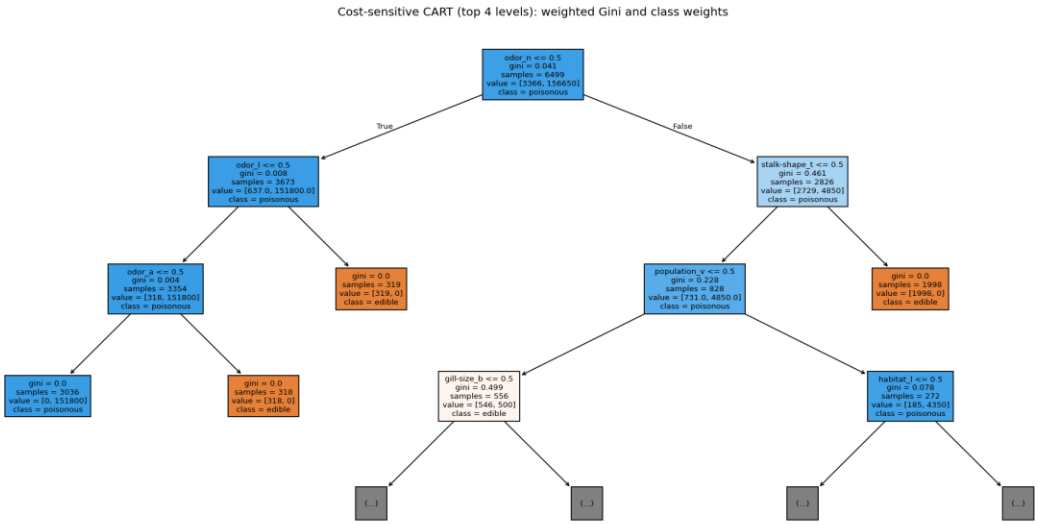


Figure 4. Cost-sensitive CART (w = 50), top 4 levels. Impurities and class values are computed with the ×50 poisonous weight (for example root Gini 0.041 and value [3366, 156650]) and are inflated; sample counts are real. The tree splits on odor first, then stalk-shape, population, gill-size and habitat.

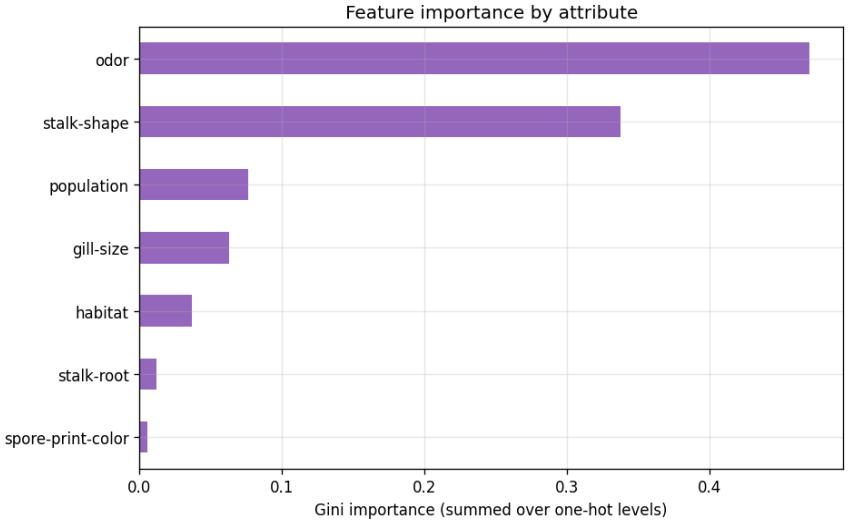


Figure 5. Feature importance (Gini decrease summed over one-hot levels) for the cost-sensitive tree: odor 0.470, stalk-shape 0.337, population 0.076, gill-size 0.063, habitat 0.037, stalk-root 0.012, spore-print-color near 0. Odor dominates in every tree fitted.

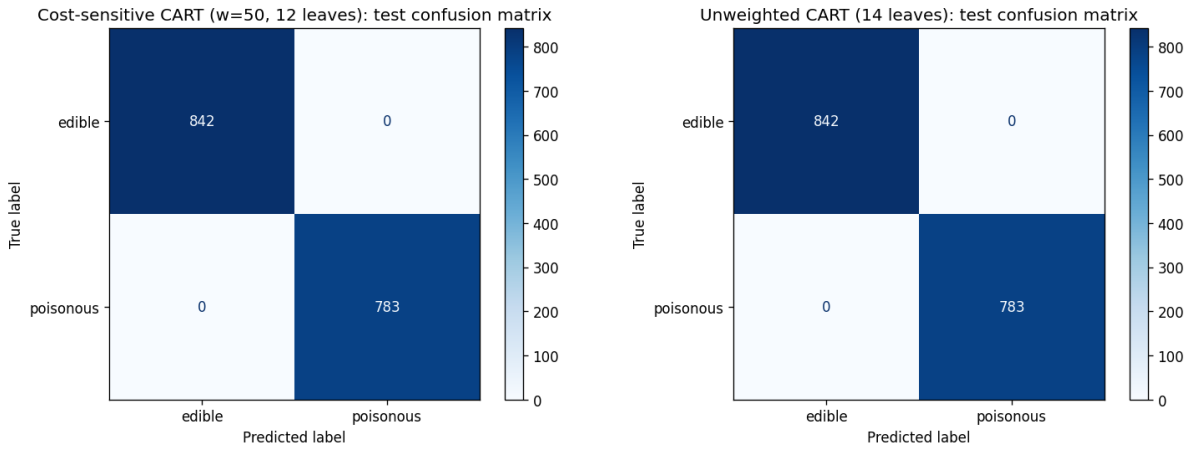


Figure 6. Test confusion matrices. Left: cost-sensitive tree (w = 50, 12 leaves). Right: unweighted CART (14 leaves). Both give 842 true edible, 783 true poisonous, 0 false negatives and 0 false positives on this split.

Section F: Error Analysis & Discussion

F.1 Where the errors come from

- **Odor identifies most poisonous specimens alone.** The rule "odor ∈ {foul, fishy, creosote, spicy, musty, pungent} ⇒ poisonous" covers 3,796 of 3,916 poisonous specimens (96.9%) with 0 edible flagged. Specimens with almond or anise odor are edible, and odor = none is mixed.
- **The remaining poisonous specimens hide in odor = none.** Among odor = none specimens, spore-print-color r gives 72 poisonous / 0 edible and w gives 48 poisonous / 576 edible; every other spore colour is edible only. The 48 poisonous white-spore specimens are the hardest cases, mixed with 576 edible ones, and need further attributes (stalk-shape, population, gill-size, habitat).
- **Unweighted trees miss about 2 poisonous specimens in cross-validation** (2 of 3,133 poisonous training rows, 0.06%). The clean unweighted test matrix is partly a property of this split. Class weighting removes the cross-validated errors, which is why the weighted tree is the safer choice.
- **Naive Bayes fails on interactions.** Its log-odds are additive over attributes, so it cannot express rules such as "odor none AND spore print white AND clustered population"; correlated attributes are also double-counted. It produces 96 false negatives (recall 87.7%) at the default threshold.

F.2 Bias–variance diagnosis

Along the pruning path, training, CV and test accuracy remain within about one point of each other, so variance is low even for the 14-leaf tree. Pruning below about 8 leaves raises bias: accuracy falls to 94.2% at 3 leaves and 88.6% at 2. Naive Bayes is high-bias (93.3% accuracy)

because of its independence assumption. Tree depth of 7 with 12–14 leaves is a good capacity point.

F.3 Feature importance is model-dependent

The brief expects odor, spore-print-color and stalk-surface to dominate. Odor does in every tree. In the unweighted tree (Figure 3), splits on stalk-root, spore-print-color and stalk-surface-below-ring appear right after the odor split, matching that expectation. In the cost-sensitive tree, stalk-shape, population, gill-size and habitat take over, and spore-print-color falls to almost zero. Correlated attributes can substitute for each other when class weights change split order, so importances should be read as "useful for this tree" and not as unique causal drivers.

F.4 Limitations

- The records are hypothetical specimens drawn from a field-guide description of about 23 species. Rows from the same species are near-duplicates, so a random split likely places near-copies of test rows in the training set. The 100% result therefore overstates real-world performance.
- Perfect accuracy on one 1,625-row split is limited evidence: only 783 poisonous specimens are in the test set, and even rare false negatives are unacceptable in a food-safety setting.
- The class weight (50) and cost ratio (100 : 1) are assumptions; a different ratio would change the selected model.

F.5 Practical recommendations

- Never use the model as the sole safety check. Use it as a screening aid alongside an expert, and discard any specimen the model or the forager is unsure about.
- Prefer the cost-sensitive tree: it matches the unweighted tree on test accuracy and removes the cross-validated false negatives. Report recalls on poisonous alongside accuracy.
- Treat "odor = none" as the danger zone: the absence of smell does not mean safe. A spore print (r or w) should be collected before any decision.
- Validate on real, independently collected specimens (split by species) before any field use, and handle unseen attribute levels explicitly.
- Do not use smell or taste tests on unknown mushrooms; the odor attribute is meant for expert description.

F.6 Reproducibility

All randomness is fixed with SEED = 42 (train/test split, cross-validation folds, tree estimators). The notebook loads the data from the UCI repository and runs top to bottom without manual steps.